

Project Title: Sign Language Recognition Using CNN

AI700-001 PROJECT PROPOSAL

Name of the Students in the group

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Name 2: Robert Cole Mullisa

Due Date:

Monday, March 2, 2026

Professor: REDA NACIF ELALAOUI

TOPIC OF INTEREST: Using CNN for Sign Language Recognition**Introduction & Motivation**

Our project is to create a CNN model that will be able to interpret images of Sign Language gestures as text. Such programs are known as SLR (Sign Language Recognition). We are going to apply Deep Learning concepts in our SLR program in order to achieve high recognition accuracy.

Problem Definition: Sign Language is an important communication tool that facilitates communication for people with hearing problems. It is not an easy language and it requires a lot of time, effort, and dedication to learn. The automation of Sign Language recognition would be a very helpful tool that would allow people without knowledge of the language to be able to understand people who use it. With the help of Deep Learning, we are going to automate this problem, so that our CNN model will analyze images of Sign Language gestures and translate them into text.

Motivation: This project will serve as a demonstration of the capabilities of a CNN model. By working on this program we will learn how to apply fundamental Deep Learning concepts in practice. We chose SLR as our project because it is a practical tool that can be used in real life to help many people. Also, such systems are relatively rare, but they are a highly researched area of AI, so we have an opportunity to make a real contribution to the development of SLR.

Input-Output Behavior: The input for our program will be a series of Sign Language gestures in the form of images spelling out a word. Upon reading the images, the program should output the corresponding text. For example, if the program is shown a series of gestures for the word “hello” then it should print out text containing the word “hello”.

Project Scope: In general, image recognition is a very complex process that relies on labeled datasets, quality of images, the presence of noise, contrast, brightness, etc. Sign Language contains gestures corresponding to individual letters, whole words, and even expressions. Sign Language is not the same between different languages but instead each spoken language has its own Sign Language. Taking into account the time constraints, we are going to limit our project to recognizing gestures corresponding to individual letters of the English alphabet.

Related Work

The three peer reviewed papers that we have studied are:

1. Hafiz Muhammad, S. K., Mcloughlin, S. D., & Murtagh, I. (2025). Comparative Evaluation and Utilization of Convolutional Neural Network Architectures for Irish Sign Language Recognition. International Journal of Combinatorial Optimization Problems and Informatics, 16(1), 123-131.

2. Taher, H. A., & Zeebaree, S. R. M. (2025). A Critical Study of Recent Deep Learning-Based Continuous Sign Language Recognition. The Review of Socionetwork Strategies, 19(1), 131-161.
3. Zhang, Y., & Jiang, X. (2024). Recent Advances on Deep Learning for Sign Language Recognition. Computer Modeling in Engineering & Sciences, 139(3), 2399-2450.

The listed papers give a comprehensive overview of Sign Languages, technologies for recognizing them, and using AI models, especially Deep Learning, for their interpretation. The papers suggest different techniques to enhance the SLR process and achieve higher recognition accuracy. The papers also list the challenges involved in Sign Language Recognition using AI models and show how recent technological achievements, including the development of Deep Learning models, are utilized to solve these issues.

Approach

Our approach for this project will be to train a CNN model on an existing data set of images representing American Sign Language gestures that correspond to letters of English alphabet. The program will use three layers of CNN to recognize each image. The Convolutional layer will use filters, padding, smoothing, etc. to preprocess an image. The Pooling layer will reduce the size of the image in order to compress the information and make the program more efficient. Finally, the Fully-connected layer will represent the Neural Network with several hidden layers that will be trained on an existing dataset and further used to recognize new images. The program will work iteratively to narrow the error margin during training and testing. Some anticipated challenges are noise in the images, poor quality of the images, uniformity of the gesture images in the dataset, etc. Some possible methods that we can use to address these challenges are choosing the best filters, using methods to eliminate noise, tuning the network parameters, such as the number of hidden layer and the number of nodes in each layer, etc.

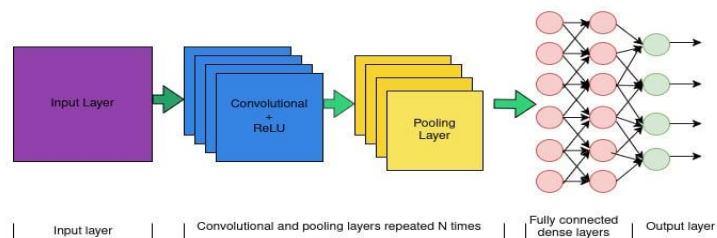


Figure 1: Example of CNN architecture [4] that we will utilize.

Source code will be using

We will be using Python source code in the VS Code environment to write our CNN model. Some of the libraries that we might use for this project are Python modules and libraries, such as Image, Plotly, Pandas, NumPy, TensorFlow, Matplotlib, OpenCV, and PyTorch. For our model we utilize the MNIST Sign Language Dataset [7]. It contains rich data of 28 x 28 images of grayscale ASL alphabet signs. The training set contains 27,455 images and labels and the testing set contains 7,172 images and labels.

Evaluation

We will evaluate our outputted result by comparing the letter predicted by our CNN model to the Sign Language gesture shown in the image. The criterion for the model performance will be accuracy. The accuracy will measure the ratio of correct matches to the total number of tested images. In addition, we will compare the accuracies produced by the training and testing stages in order to check for underfitting and overfitting. We are going to tune the CNN parameters and the number of epochs to achieve the best accuracy and we expect the accuracy to increase as we increase the volume of training data. We plan to achieve an accuracy of at least 90% or higher.

Milestones/Timelines

The milestones with their corresponding expected dates for achievement:

Dates (Weeks)	Describe the Objectives and roadmap actions	Contributor(s)
Week 6	Topic selection and proposal submission	Michael, Robert
Week 7	Create a repository on GitHub	Robert
Week 8	Data acquisition and pre-processing	Michael
Week 9	Create structure for CNN model	Robert
Week 10	Report project progress	Michael, Robert
Week 11	Implement model using chosen source code	Michael
Week 12	Train/test split	Robert
Week 13	Evaluate and improve model based on test results	Michael
Week 14	Repeat testing and improvement cycle	Robert
Week 15	The final presentation of project results & discussion	Michael, Robert

References

1. Abdullah, A., Amoudi, G. A., & Alghamdi, H. S. (2024). Advancements in Sign Language Recognition: A Comprehensive Review and Future Prospects. *IEEE Access*, 12, 128871–128895.
2. Alam, I., Hameed, A., & Riaz, A. Z. (2024). Exploring Sign Language Detection on Smartphones: A Systematic Review of Machine and Deep Learning Approaches. *Advances in Human - Computer Interaction*, 2024.
3. Ankita, W., & Parteek, K. (2021). Sign Language Recognition Systems: A Decade Systematic Literature Review. *Archives of Computational Methods in Engineering*, 28(3), 785-813.
4. Codecademy. (2025, March 27). Understanding Convolutional Neural Network (CNN) Architecture. Codecademy. <https://www.codecademy.com/article/understanding-convolutional-neural-network-cnn-architecture>.
5. Nagagopiraju, V., Reddy, Y. V., Raju, C. S., Sai, N. S., & Vishnu, N. (2025). Deep Learning and Computer Vision Based Sign Language Dynamic Gesture Recognition System. *International Journal of Communication Networks and Information Security*, 17(3), 89-95.
6. Rastgoo, R., Kiani, K., & Escalera, S. (2021). Sign Language Recognition: A Deep Survey. *Expert Systems with Applications*, 164, 1.
7. Sign Language MNIST. (n.d.). *Www.kaggle.com*. <https://www.kaggle.com/datasets/datamunge/sign-language-mnist>
8. Varma, D. M., & Ch. Sekhar. (2024). Advances in AI for Automatic Sign Language Recognition: A Comparative Study of Machine Learning Approaches. *I-Manager's Journal on Software Engineering*, 19(2), 10.